

Rita Sevastjanova

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Senior Applied Scientist with 10+ years of experience at the intersection of cutting-edge research and real-world deployment of artificial intelligence (AI). Proven track record in leading high-impact research programs, securing competitive funding, and driving industry collaborations and revenue generation. Recognized for advancing the understanding of model behavior and fairness, grounded in award-winning doctoral research on the linguistic capabilities and limitations of large language models (LLMs). Currently focused on explainability of LLMs and multimodal models at ETH Zurich, developing methods to mitigate hallucinations, resolve ambiguous user queries, and enable safe, trustworthy deployment of agent-based systems in production environments.

EXPERIENCE

ETH ZURICH

POSTDOCTORAL RESEARCHER IN COMPUTER SCIENCE
Interactive Visualization and Intelligence Augmentation Lab
Nov 2023 – now | Zurich, Switzerland

UNIVERSITY OF KONSTANZ

RESEARCH ASSOCIATE IN COMPUTER SCIENCE
Data Analysis and Visualization Group
Nov 2017 – Nov 2023 | Konstanz, Germany

UNIVERSITY OF KONSTANZ

RESEARCH ASSISTANT IN COMPUTER SCIENCE
Data Analysis and Visualization Group
Aug 2013 – Sept 2017 | Konstanz, Germany

PROJECTS

ARTEFACTS OF THE FUTURE | WITH BIOETHICS LAB, ETHZ |

PROJECT LEAD AND DEVELOPER
Oct 2025 – Present | Zurich, Switzerland
The project is a speculative exhibition that pairs 3D-printed sculptures with AI-powered, RAG-based chatbots, each tailored to the unique persona of its corresponding artefact.

PERSONALIZED VA | WITH IVDA LAB, UZH | PROJECT LEAD

AND DEVELOPER
Sept 2025 – Present | Zurich, Switzerland
The project explores natural language as a medium for eliciting user preferences in ranking-based tasks, while addressing ambiguity and uncertainty in interpretation and modeling.

ELEMENTS OF APN | WITH ZHDK | PROJECT LEAD

2024 – 2025 | Zurich, Switzerland
The project applied similarity metrics and explainable AI methods (e.g., saliency maps) for interactively comparing visual artworks.

REPORT ANALYSIS | FUNDED BY GIZ | DEVELOPER

2022 – 2023 | Konstanz, Germany
The project combined LLM fine-tuning and text mining methods for analyzing evaluation reports of international organizations.

LINVIS.IO | DEVELOPER

2017 – 2023 | Konstanz, Germany
The project developed text-mining pipelines and web-based visualizations for applications ranging from (e.g., political) discourse analysis to language model explainability.

QUESTIONS VISUALIZED | FUNDED BY DFG | DEVELOPER

2017 – 2023 | Konstanz, Germany
The project conducted interdisciplinary research on question classification tasks and developed interactive visual analytics systems for question annotation and analysis.

PERSONAL INFORMATION

DATE OF BIRTH: 27.09.1988

NATIONALITY: LATVIAN

EDUCATION

UNIVERSITY OF KONSTANZ

P.H.D. IN NATURAL SCIENCES (DR.RER.NAT.)
2017-2023 | Konstanz, Germany

Thesis: "Interactive Visual Investigation of Word Embedding Contextualizations in Large Language Models"

Received Südwestmetall-Förderpreis (5000€) for an outstanding dissertation.

UNIVERSITY OF KONSTANZ

MASTER OF SCIENCE, COMPUTER AND INFORMATION SCIENCE
2016-2017 | Konstanz, Germany

UNIVERSITY OF KONSTANZ

BACHELOR OF SCIENCE, INFORMATION ENGINEERING
2012-2016 | Konstanz, Germany

UNIVERSITY OF LATVIA

BACHELOR OF ARTS, SINOLOGY
2008-2011 | Riga, Latvia

RESPONSIBILITIES

ACQUISITION OF FUNDING

Nov 2023 – now | Zurich, Switzerland
Leads the development of research proposals and the acquisition of external research funding.

RECRUITMENT

Nov 2023 – now | Zurich, Switzerland
Leads recruitment of PhD candidates and evaluation of fellowship applications.

PROJECT MANAGEMENT

Nov 2023 – now | Zurich, Switzerland
Develops and manages collaborations with external partners, negotiating project scope and aligning objectives.

SUPERVISION

Nov 2023 – now | Zurich, Switzerland
Leads a focus group of PhD students, supporting professional development and driving project progress. Supervised 15+ Bachelor's and Master's theses.

VISARGUE | FUNDED BY DFG | DEVELOPER

2013 – 2017 | Konstanz, Germany

The project developed text-mining pipelines and web-based visualizations for political discourse analysis.

INVITED TALKS AND PANEL DISCUSSIONS

COLLEGIUM HELVETICUM

Jan 2025 | Zurich, Switzerland

Panel and talk "Visual Analytics Approaches for Exploring LLMs"

IPAI EXPERIENCE DAY

Oct 2024 | Heilbronn, Germany

Talk „Investigating Biases in AI through Interactive Machine Learning“

DIGITALGIPFEL DER BUNDESREGIERUNG

Oct 2024 | Frankfurt am Main, Germany

Panel „KI im Tagesgeschäft? – Erfolgsrezepte aus Unternehmen, öffentlicher Verwaltung und Politik“

TED AI

Oct 2024 | Vienna, Austria

Co-organized the workshop "Discovering the Power of Human-AI Collaboration Firsthand"

AI + X SUMMIT

Oct 2024 | Zurich, Switzerland

Talk „Investigating Biases in AI through Interactive Machine Learning“

GOOGLE CLOUD PARTNER CONNECT

Sept 2024 | Zurich, Switzerland

Talk „Bias in Generative Language Models“

COLLECTIVE GATHERING AT THE STARTUP DAYS

May 2024 | Bern, Switzerland

Panel „Empathy in AI“

SELECTED PUBLICATIONS

FIRST-AUTHORED

- **Rita Sevastjanova**, Robin Gerling, Thilo Spinner, and Mennatallah El-Assady. 'LayerFlow: Layer-wise Exploration of LLM Embeddings using Uncertainty-aware Interlinked Projections.' In Computer Graphics Forum, 44: e70123, (2025).
- **Rita Sevastjanova**, Eren Cakmak, Shauli Ravfogel, Ryan Cotterell, and Mennatallah El-Assady. 'Visual Comparison of Language Model Adaptation.' IEEE Trans. on Visualization and Computer Graphics, 29(1): 1178--1188, (2022).
- Aikaterini-Lida Kalouli*, **Rita Sevastjanova***, Christin Beck, and Maribel Romero. 'Negation, Coordination, and Quantifiers in Contextualized Language Models.' In Proc. of the 29th Int. Conf. on Computational Linguistics (COLING), pp. 3074--3085, (2022). * Equal Contribution.
- **Rita Sevastjanova**, Aikaterini-Lida Kalouli, Christin Beck, Hanna Schäfer, and Mennatallah El-Assady. 'Explaining Contextualization in Language Models using Visual Analytics.' In: Proc. of the 59th Annual Meeting of the Association for Computational Linguistics (ACL) and the 11th Int. Joint Conf. on Natural Language Processing (2021), pp. 464–476.

CLOSELY-SUPERVISED

- T. Stähle, P. F. Gyarmati, T. Spinner, **R. Sevastjanova**, D. Moritz, and M. El-Assady. 'VACP: Visual Analytics Context Protocol.' arXiv preprint arXiv:2603.29322, (2026), Currently Under Review.
- Robin Shing Moon Chan, **Rita Sevastjanova**, and Mennatallah El-Assady. 'PleaSQLarify: Visual Pragmatic Repair for Natural Language Database Querying.' In Proc. of the 2026 CHI Conf. on Human Factors in Computing Systems (CHI), Article No.: 1163, pp. 1--18, (2026). DOI: 10.1145/3772318.3791265. ★ Received Best Paper Award.
- Thilo Spinner, **Rita Sevastjanova**, Rebecca Kehlbeck, Tobias Stähle, Daniel A. Keim, Oliver Deussen, Andreas Spitz, and Mennatallah El-Assady. 'Revealing the Unwritten: Visual Investigation of Beam Search Trees to Address Language Model Prompting Challenges.' In Proc. of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL), System Demonstrations, pages 295–306, (2025).
- Kenza Amara, **Rita Sevastjanova**, and Mennatallah El-Assady. 'SyntaxShap: Syntax-aware Explainability Method for Text Generation.' In Findings of the Association for Computational Linguistics (ACL) 2024, pages 4551–4566, (2024).

AWARDS

- 2026 Best Paper Award, CHI Conference on Human Factors in Computing Systems
 - 2025 Südwestmetall-Förderpreis for an Outstanding Dissertation
 - 2021 Outstanding Paper Award, Int. Conf. on Computational Semantics
 - 2021 Best Paper Award, Journal of Comparative Policy Analysis
 - 2017 Honorable Mention Paper Award, VAST Conference
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SKILLS

PROGRAMMING

Python • Java • JavaScript • TypeScript • HTML • CSS • D3.js • SQL

FRAMEWORKS/PLATFORMS

TensorFlow • PyTorch • Scikit-learn • React • Node.js • Git • Docker

LANGUAGES

Latvian (native) • English (proficient) • German (proficient) • Russian (fluent) • Chinese (basic)

PUBLICATIONS

50+ (co-)authored peer-reviewed publications at top-tier venues such as IEEE VIS, EuroVis, CHI, ACL, COLING.
